Margin-stable schemes and the Baire category of full complexity in non-autonomous dynamics on the Hilbert cube

Yitzchak Shmalo
Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel
yitzchak.shmalo@gmail.com

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Abstract

In the full non-autonomous mapping space $\mathcal{F}(Q_\infty) = C(Q_\infty, Q_\infty)^{\mathbb{N}}$ of the Hilbert cube, equipped with the uniform-in-time metric, the recent construction of [1] produces a *dense* set of systems admitting exact windowed Hilbert horseshoes, with all-tail symbolic semifactors, windowed distributional chaos of type 1, infinite topological entropy, and infinite metric mean dimension; the same paper observes that the exact class is not stable and asks (Section 14 of [1]) for a residual theorem obtained by a "Baire-fusion mechanism." This paper resolves the stability side of that question. First, we introduce *strict depth-N brick schemes*, margin-protected finite-coordinate covering configurations, and prove that for every depth N and every sequence of dense open windows the corresponding class $\mathcal{U}_N$ is open and dense, with explicit stability radius $\theta\, 2^{-N-1}$ for a scheme of margin θ. Consequently the class $\mathcal{S} = \bigcap_N \mathcal{U}_N$ of systems carrying windowed strict schemes of *every* finite depth is residual. Second, on $\mathcal{S}$ every branch brick covers every depth-N coordinate pattern of the next bricks *exactly*, by a Poincaré–Miranda argument; this is the maximal one-step covering compatible with stability. Third, a genuinely dynamical statement holds residually: for a residual set of systems, every tail exhibits separated orbit growth of every order q at arbitrarily fine scales ("entropy bursts"), the residual shadow of the dense infinite-metric-mean-dimension theorem. Fourth, we prove an obstruction theorem: the exact-horseshoe class of [1] has *empty interior*, and indeed every property that forces some image to have full range in arbitrarily deep coordinates has empty interior. The obstruction shows that the depth stratification is not an artifact: no margin-stable strengthening of the exact theorem can be open, so the residual hierarchy $\mathcal{S}$ is the natural generic object. We close with a precise formulation of what remains open.

1 Introduction

A non-autonomous discrete dynamical system on a compact metric space (X, ρ) is a sequence $f = (f_1, f_2, \dots)$ of continuous self-maps of X, acting through the compositions $f_n \circ \cdots \circ f_1$. The full mapping space

$$\mathcal{F}(X) = C(X, X)^{\mathbb{N}}, \qquad D_X(f, g) = \sup_{n \geq 1} \sup_{x \in X} \rho\big(f_n(x), g_n(x)\big),$$

is a complete, hence Baire, metric space (Lemma 2.1 below). For interval systems and for systems on finite-dimensional cubes and compact manifolds, the paper [1] constructs D-open dense sets of systems displaying *all-tail full complexity*: after deleting any finite initial segment, the remaining system admits a compact symbolic semifactor onto a full variable-alphabet shift,

a continuum-sized scrambled set for distributional chaos of type 1 visiting prescribed dense open windows at prescribed absolute times, infinite topological entropy, and the sharp small-scale entropy spectrum $H_{g,r}(\varepsilon) = d\log(1/\varepsilon) + O(1)$. The proofs rest on *strict* cubical covering relations: face inequalities with a positive margin θ, which survive every perturbation of size less than θ and are converted into covering statements by the Poincaré–Miranda theorem [2, 3].

On the Hilbert cube

$$Q_\infty = [0,1]^{\mathbb{N}}, \qquad \rho(x,y) = \sum_{j=1}^{\infty} 2^{-j}\,|x_j - y_j|,$$

the situation changes. The result of [1] on Q_∞ is a *dense* theorem: a dense set of systems admits exact windowed Hilbert horseshoes (Definition 7.1 below), built from finite-support boxes mapped *exactly* onto the next box. Exactness in arbitrarily high coordinates is not stable, because coordinate j carries metric weight 2^{-j}: a perturbation of uniform size δ can move the j-th image coordinate by $2^j\delta$. Section 14 of [1] records the resulting question, which is the starting point of the present paper:

> "The Hilbert-cube conclusion is dense but is not asserted to be open: exact covering in arbitrarily high coordinates is not stable under the Hilbert metric. A residual Hilbert-cube theorem with coherent all-tail symbolic codings would require an additional Baire-fusion mechanism beyond the exact finite-support implantations proved here."

This paper carries out the stability analysis that any such fusion mechanism must confront, and proves four results: a residual depth hierarchy of margin-stable schemes (Theorem 1.1), exact one-step covering at every depth on that residual set (Theorem 1.2), a residual entropy-burst theorem in every tail (Theorem 6.2; see Section 6), and an empty-interior obstruction (Theorem 1.3).

Main results

Fix a sequence $W_\bullet = (W_n)_{n\ge1}$ of dense open subsets of Q_∞ (*windows*) and geometric data $(M_n, q_n, \gamma_n)_{n\ge1}$ as in Section 3 (finitely many branches $M_n \ge 2$, separation dimensions $q_n \ge 1$, diameter bounds $\gamma_n > 0$; the block schedule of [1, Section 12] is the motivating instance). For each integer $N \ge 1$ we define in Section 3 the class

$$\mathcal{U}_N(W_\bullet) \subseteq \mathcal{F}(Q_\infty)$$

of systems admitting a *strict depth-N windowed brick scheme*: a time-indexed family of pairwise separated finite-support *bricks* inside the windows, such that every brick crosses, in every coordinate $\ell \le N$ and with a uniform margin $\theta > 0$, the coordinate range of all bricks at the next time.

Theorem 1.1 (Finite-depth genericity; Theorem 5.1 below)**.** *For every depth $N \ge 1$ and every window sequence $W_\bullet$, the class $\mathcal{U}_N(W_\bullet)$ is open and dense in $(\mathcal{F}(Q_\infty), D)$. A scheme with margin θ persists, with margin $\theta/2$, on the entire ball of radius $\theta\,2^{-N-1}$ around the realizing system. Consequently*

$$\mathcal{S}(W_\bullet) \;=\; \bigcap_{N\ge1} \mathcal{U}_N(W_\bullet)$$

is residual in $\mathcal{F}(Q_\infty)$, as is the simultaneous class $\bigcap_{a\geq 1}\mathcal{S}(W_\bullet^a)$ for any countable family of window sequences.

Theorem 1.2 (Generic exact one-step covering; Theorem 3.4 below)**.** *Let $g \in \mathcal{S}(W_\bullet)$. Then for every $N \geq 1$ there is a depth-N scheme for g such that for every time n, every brick $B_{n,i}$ of the scheme, and every target pattern $t = (t_\ell)_{\ell\leq N}$ lying coordinatewise in the interval spanned by the bricks at time $n+1$, there exists $x \in B_{n,i}$ with*

$$\pi_\ell\, g_n(x) = t_\ell \qquad (1 \leq \ell \leq N).$$

In particular every brick maps exactly onto every depth-N coordinate pattern of every brick at the next time, inside the prescribed windows.

Theorem 1.3 (Obstruction: the exact class has empty interior; Theorem 7.2 below)**.** *Let $\mathsf{Ex} \subseteq \mathcal{F}(Q_\infty)$ denote the set of systems admitting an exact windowed Hilbert horseshoe in the sense of [1, Definition 12.1] (Definition 7.1 below). Then Ex is dense and has empty interior. More generally, every class $\mathcal{P} \subseteq \mathcal{F}(Q_\infty)$ with the property that each $g \in \mathcal{P}$ has, for some n and some compact $A \subseteq Q_\infty$, an image $g_n(A)$ whose coordinate projections are onto $[0,1]$ in infinitely many coordinates, has empty interior.*

Theorem 1.3 shows that the "dense, not open" phenomenon of [1] is intrinsic and not an artifact of the construction: *no* sharpening of exact infinite-depth covering can be open, since arbitrarily small uniform perturbations (an explicit coordinate squeeze) destroy full range in deep coordinates everywhere in the space. Theorems 1.1 and 1.2 identify the optimal margin-stable substitute: full strict covering of every finite depth, simultaneously for all depths, on a residual set. The stability radius $\theta 2^{-N-1}$ quantifies the cost of depth and explains the stratification: the margins of conditions at depth N have stability comparable to 2^{-N}, so no single open set can certify unbounded depth, while the intersection over N retains residuality.

Discussion: what the fusion mechanism can and cannot give

It is natural to ask whether membership in $\mathcal{S}(W_\bullet)$ already implies the *dynamical* conclusions of the dense theorem of [1] (semifactors, DC1, infinite entropy and metric mean dimension in every tail). We do not claim this, and we believe the question is genuinely delicate, for a reason worth recording. The infinite-horizon conclusions all require orbits threading prescribed bricks at *all* times. The backward-induction production of such orbits in [1] consumes exact surjectivity onto the next box in *all* coordinates; a depth-N scheme controls only N image coordinates per step, and the uncontrolled deep coordinates of the image re-enter as uncontrolled *inputs* at the next step. Quantitatively, a condition involving image coordinate ℓ has stability $O(2^{-\ell})$ under D, while restoring such a condition after a perturbation of size r costs $\Omega(\min(1, 2^\ell r))$ per coordinate in amplitude; summing over the critical band of coordinates, maintenance of unbounded-depth margins along a Baire fusion fails against an adversary that shrinks neighbourhoods faster than geometrically. This is exactly the obstruction isolated in Theorem 1.3 in its sharpest open form. We therefore state the remaining problem precisely in Section 8, in a form that separates what is now known (the residual depth hierarchy, with exact one-step covering at every depth, plus the empty-interior obstruction) from what is open (residuality of the infinite-horizon conclusions themselves).

Relation to the finite-dimensional theory

On $Q_d = [0,1]^d$ with the sup metric, all d coordinates have unit weight, margins have uniform stability, and [1] obtains a single open dense class with the full dynamical conclusions. The present results show in what precise sense $d = \infty$ differs: openness survives at every finite depth with an exponentially small but positive radius, and fails at infinite depth for structural reasons. The residual set $\mathcal{S}(W_\bullet)$ is the direct analogue of the finite-dimensional open dense class under this constraint.

Organization

Section 2 fixes notation and records the Baire property. Section 3 defines strict depth-N brick schemes and proves openness and the covering theorem. Section 4 proves the selection and implantation lemmas and density. Section 5 assembles Theorem 1.1. Section 6 proves the residual entropy-burst theorem. Section 7 proves the obstruction theorem. Section 8 states the remaining open problem.

2 Preliminaries

Throughout, $Q_\infty = [0,1]^{\mathbb{N}}$ carries the metric $\rho(x,y) = \sum_{j\ge1} 2^{-j}|x_j - y_j|$, which induces the product topology and makes Q_∞ compact. We write $\pi_\ell : Q_\infty \to [0,1]$ for the ℓ-th coordinate projection. The mapping space $C(Q_\infty, Q_\infty)$ carries the uniform metric $d_\infty(u,v) = \sup_x \rho(u(x), v(x))$, and

$$\mathcal{F}(Q_\infty) = C(Q_\infty, Q_\infty)^{\mathbb{N}}, \qquad D(f,g) = \sup_{n\ge1} d_\infty(f_n, g_n).$$

Lemma 2.1 ([1, Lemma 2.1])**.** *$(\mathcal{F}(Q_\infty), D)$ is a complete metric space, and hence a Baire space.*

Proof. $C(Q_\infty, Q_\infty)$ is complete because Q_∞ is compact and uniform limits of continuous maps are continuous. If $(f^{(k)})_k$ is D-Cauchy, each coordinate sequence $(f_n^{(k)})_k$ is d_∞-Cauchy with a modulus independent of n; the coordinate limits f_n assemble to $f \in \mathcal{F}(Q_\infty)$ with $D(f^{(k)}, f) \to 0$. Completeness implies the Baire property. □

We use the elementary weight estimate repeatedly: for $u, v \in C(Q_\infty, Q_\infty)$ and any coordinate ℓ,

$$\sup_{x\in Q_\infty} \big|\pi_\ell u(x) - \pi_\ell v(x)\big| \;\le\; 2^\ell\, d_\infty(u,v), \qquad d_\infty(u,v) \;\le\; \sum_{\ell\ge1} 2^{-\ell} \sup_x |\pi_\ell u(x) - \pi_\ell v(x)|. \tag{1}$$

A *finite-support box* is a product $B = \prod_{j\ge1} I_j$ with each $I_j \subseteq [0,1]$ a nondegenerate closed interval and $I_j = [0,1]$ for all $j > R$; the least such R is the *support level*. Finite-support boxes are compact with nonempty interior, and they form a local base: every nonempty open $U \subseteq Q_\infty$ contains finite-support boxes of arbitrarily small diameter [1, Lemma 10.2]. For $K \ge 1$ and $\alpha \in (0,1)$, an (α, K)-*box* is a finite-support box of the form $J = \prod_{j=1}^K [a_j, a_j + \alpha] \times [0,1]^{\{K+1, K+2, \dots\}}$; then $\operatorname{diam}_\rho(J) \le \alpha + 2^{-K}$.

Finally we recall the Poincaré–Miranda theorem [2, 3].

Theorem 2.2 (Poincaré–Miranda)**.** *Let $P = \prod_{\ell=1}^N [r_\ell, s_\ell] \subset \mathbb{R}^N$ and let $\Phi = (\Phi_1, \dots, \Phi_N) : P \to \mathbb{R}^N$ be continuous. If for every ℓ, $\Phi_\ell \le 0$ on $\{z \in P : z_\ell = r_\ell\}$ and $\Phi_\ell \ge 0$ on $\{z \in P : z_\ell = s_\ell\}$, then Φ has a zero in P.*

3 Strict depth-N brick schemes

Fix once and for all *geometric data*: integers $M_n \geq 2$ (branch counts), $q_n \geq 1$ (separation dimensions), and reals $\gamma_n > 0$ (diameter bounds), $n \geq 1$. For the dynamical applications of [1] one takes the universal block schedule of [1, Section 12], where every label C_a, D_s, $E_{q,m}$ occurs on infinitely many blocks and $\gamma_n \to 0$; none of the results below depends on that specific choice. Fix also a sequence $W_\bullet = (W_n)$ of dense open subsets of Q_∞.

Definition 3.1 (Bricks)**.** *A* brick *with support level R is a finite-support box $B = \prod_{j \leq R} I_j \times [0,1]^{\{R+1,\dots\}}$ with all restricted intervals I_j nondegenerate. For bricks B, B' and $q \geq 1$ we write*

$$\operatorname{dist}_q(B, B') = \inf_{x \in B,\, y \in B'} \max_{1 \leq \ell \leq q} |x_\ell - y_\ell|.$$

Definition 3.2 (Strict depth-N windowed brick scheme)**.** *Let $N \geq 1$. A system $g \in \mathcal{F}(Q_\infty)$* admits a strict depth-$N$ $W_\bullet$-windowed brick scheme *if there exist $\alpha \in (0, 1/4)$, a margin $\theta \in (0, 1/16]$, support levels $R_n \geq \max(N, q_n)$, and bricks*

$$B_{n,1}, \dots, B_{n,M_n} \subseteq W_n, \qquad B_{n,i} = \prod_{j \leq R_n} I_j^{n,i} \times [0,1]^{\{R_n+1,\dots\}},$$

such that, writing

$$m^-_{n,\ell} = \min_{1 \leq i \leq M_n} \min I_\ell^{n,i}, \qquad m^+_{n,\ell} = \max_{1 \leq i \leq M_n} \max I_\ell^{n,i} \qquad (\ell \leq N),$$

the following hold for all $n \geq 1$:

(S0) (anchoring) *$I_\ell^{n,i} \subseteq [2\theta,\, 1 - 2\theta]$ for all i and all $\ell \leq N$;*

(S1) $\operatorname{diam}_\rho(B_{n,i}) < \gamma_n$ *for all i;*

(S2) $\operatorname{dist}_{q_n}(B_{n,i}, B_{n,j}) > c_{\alpha,q_n} M_n^{-1/q_n}$ *for $i \neq j$, where $c_{\alpha,q} > 0$ are the selection constants of Lemma 4.2;*

(S3) (strict depth-N crossing) *for every i and every $\ell \leq N$,*

$$\pi_\ell\, g_n(x) \;\leq\; m^-_{n+1,\ell} - \theta \quad \text{for all } x \in B_{n,i} \text{ with } x_\ell = \min I_\ell^{n,i},$$

$$\pi_\ell\, g_n(x) \;\geq\; m^+_{n+1,\ell} + \theta \quad \text{for all } x \in B_{n,i} \text{ with } x_\ell = \max I_\ell^{n,i}.$$

We write $\mathcal{U}_N(W_\bullet)$ for the set of systems admitting such a scheme. By (S0), $m^-_{n+1,\ell} - \theta \geq \theta > 0$ and $m^+_{n+1,\ell} + \theta \leq 1 - \theta < 1$, so the inequalities in (S3) are never vacuous for range reasons.

Lemma 3.3 (Openness with quantified radius)**.** *Let $g \in \mathcal{U}_N(W_\bullet)$, realized by a scheme with margin θ. If $h \in \mathcal{F}(Q_\infty)$ satisfies $D(h, g) < \theta\, 2^{-N-1}$, then h realizes the* same *bricks as a strict depth-N scheme with margin $\theta/2$. In particular $\mathcal{U}_N(W_\bullet)$ is open in $(\mathcal{F}(Q_\infty), D)$.*

Proof. Conditions (S1) and (S2) and the window membership concern only the bricks. For (S3), fix $n, i, \ell \leq N$ and $x \in B_{n,i}$ with $x_\ell = \min I_\ell^{n,i}$. By (1),

$$\pi_\ell h_n(x) \;\leq\; \pi_\ell g_n(x) + 2^\ell D(h, g) \;<\; \big(m^-_{n+1,\ell} - \theta\big) + 2^N\, \theta\, 2^{-N-1} \;=\; m^-_{n+1,\ell} - \tfrac{\theta}{2},$$

and symmetrically on the faces $x_\ell = \max I_\ell^{n,i}$. □

Theorem 3.4 (Exact one-step depth-N covering)**.** *Let g admit a strict depth-N scheme with bricks $B_{n,i}$ and margin θ. Then for every $n \geq 1$, every $1 \leq i \leq M_n$, and every target pattern $t = (t_\ell)_{\ell \leq N}$ with $t_\ell \in [m^-_{n+1,\ell},\, m^+_{n+1,\ell}]$ for all $\ell \leq N$, there exists $x \in B_{n,i}$ with*

$$\pi_\ell\, g_n(x) = t_\ell \qquad (1 \leq \ell \leq N).$$

In particular, for every $1 \leq i' \leq M_{n+1}$ there is $x \in B_{n,i}$ whose image $g_n(x)$ agrees in its first N coordinates with a prescribed point of $B_{n+1,i'}$; all such x lie in $B_{n,i} \subseteq W_n$.

Proof. Write $I_\ell = I_\ell^{n,i}$. Define the section $\sigma : \prod_{\ell \leq N} I_\ell \to B_{n,i}$ by $\sigma(z)_\ell = z_\ell$ for $\ell \leq N$, $\sigma(z)_j = \operatorname{mid} I_j^{n,i}$ for $N < j \leq R_n$, and $\sigma(z)_j = 1/2$ for $j > R_n$; σ is continuous and takes values in $B_{n,i}$. Let

$$\Phi_\ell(z) = \pi_\ell\, g_n(\sigma(z)) - t_\ell \qquad (\ell \leq N).$$

If $z_\ell = \min I_\ell$, then $\sigma(z) \in B_{n,i}$ has ℓ-th coordinate $\min I_\ell$, so by (S3), $\Phi_\ell(z) \leq m^-_{n+1,\ell} - \theta - t_\ell \leq -\theta < 0$; if $z_\ell = \max I_\ell$, then $\Phi_\ell(z) \geq \theta > 0$. By Theorem 2.2 there is z with $\Phi(z) = 0$, and $x = \sigma(z)$ is the required point. For the final assertion take $t_\ell \in I_\ell^{n+1,i'} \subseteq [m^-_{n+1,\ell}, m^+_{n+1,\ell}]$ for $\ell \leq N$. □

Remark 3.5. *The radius $\theta 2^{-N-1}$ in Lemma 3.3 is the correct order of the stability of depth-N information: a perturbation supported in the N-th image coordinate of size 2θ in amplitude has D-size $2^{-N+1}\theta$ and destroys the margin-θ inequalities (S3) at $\ell = N$ for the given scheme. Stability of margin conditions thus decays exponentially in depth, which is the quantitative form of the instability observed in [1, Section 14] and the reason the classes $\mathcal{U}_N$ are stratified by depth rather than united in a single open class; Theorem 7.2 below makes the impossibility of the united class precise.*

4 Selection, implantation, and density

We first record the measure-theoretic separation lemma and the windowed selection lemma of [1], in the form needed here.

Lemma 4.1 (Separated points; [1, Lemma 5.1])**.** *Let $A \subseteq \mathbb{R}^q$ be bounded and measurable with $\lambda_q(A) > 0$. For every $m \geq 2$ the set A contains m points whose pairwise sup distances exceed $\frac{1}{4}\,\lambda_q(A)^{1/q} m^{-1/q}$.*

Proof. Put $r = \frac{1}{16}(\lambda_q(A)/m)^{1/q}$ and let $P \subseteq A$ be a maximal $4r$-separated subset. If $\operatorname{card}(P) \leq m-1$, maximality lets A be covered by at most $m-1$ sup-balls of radius $4r$, whence $\lambda_q(A) \leq (m-1)(8r)^q < 2^{-q}\lambda_q(A)$, a contradiction. □

Lemma 4.2 (Windowed selection)**.** *Fix $\alpha \in (0, 1/4)$ and $q \geq 1$. There is a constant $c_{\alpha,q} > 0$ with the following property. Let $J = \prod_{j \leq K}[u_j, u_j + \alpha] \times [0,1]^{\{K+1,\dots\}}$ be an (α, K)-box with $K \geq q$, let $W \subseteq Q_\infty$ be dense open, let $f \in C(Q_\infty, Q_\infty)$, and let $m \geq 2$. Then there are points $p_1, \dots, p_m \in \operatorname{int}(J) \cap W$ such that*

$$\max_{1 \leq \ell \leq q} \bigl|(p_i)_\ell - (p_j)_\ell\bigr| > 4c_{\alpha,q}\, m^{-1/q} \quad (i \neq j), \qquad \operatorname{diam}_\rho\{f(p_1), \dots, f(p_m)\} < \alpha/16.$$

Proof. Define a slice $\iota : P \to \mathrm{int}(J)$, $P = \prod_{\ell \le q}[u_\ell + \alpha/4,\, u_\ell + 3\alpha/4]$, by $(\iota z)_\ell = z_\ell$ for $\ell \le q$, $(\iota z)_j = u_j + \alpha/2$ for $q < j \le K$, and $(\iota z)_j = 1/2$ for $j > K$. By compactness cover Q_∞ by Borel sets $R_1, \dots, R_P$ of ρ-diameter less than $\alpha/64$, where $P = P_\alpha$ depends only on α. The sets $A_s = \{z \in P : f(\iota z) \in R_s\}$ cover P, so some A_s has $\lambda_q(A_s) \ge (\alpha/2)^q P_\alpha^{-1}$. Lemma 4.1 gives $z_1, \dots, z_m \in A_s$ pairwise more than $8c_{\alpha,q}m^{-1/q}$ apart in sup norm, where $c_{\alpha,q} = \frac{1}{32}\big((\alpha/2)^q P_\alpha^{-1}\big)^{1/q}$. All $f(\iota z_i)$ lie in one R_s. By continuity of f, choose pairwise disjoint relatively open $U_i \ni \iota z_i$ inside $\mathrm{int}(J)$ with first-q separation above $4c_{\alpha,q}m^{-1/q}$ between different U_i and $\rho(f(u), f(\iota z_i)) < \alpha/64$ on U_i, and pick $p_i \in U_i \cap W$ by density of W. □

Lemma 4.3 (Implantation; density of $\mathcal{U}_N(W_\bullet)$)**.** *For every $N \ge 1$, every $f \in \mathcal{F}(Q_\infty)$, and every $\eta > 0$ there exists $g \in \mathcal{U}_N(W_\bullet)$ with $D(f, g) < \eta$.*

Proof. Set $\alpha = \min(\eta/8, 1/8)$ and $\theta = \alpha/8 \le 1/64$. Let $K_n = \max\big(N, q_n, \lceil \log_2(1/\alpha) \rceil + 1\big)$. All sites below are (α, K_n)-boxes *anchored* in the sense that their restricted intervals lie in $[2\theta,\, 1 - 2\theta]$.

Let J_1 be any anchored (α, K_1)-box. Assume the anchored site J_n is defined. Apply Lemma 4.2 to $(J_n, W_n, f_n, q_n, M_n)$ to obtain points $p_{n,1}, \dots, p_{n,M_n} \in \mathrm{int}(J_n) \cap W_n$ with first-q_n pairwise separation above $4c_{\alpha,q_n}M_n^{-1/q_n}$ and f_n-image diameter below $\alpha/16$. Choose pairwise disjoint open sets $O_{n,i} \subseteq \mathrm{int}(J_n) \cap W_n$ with $p_{n,i} \in O_{n,i}$, first-q_n separation above $c_{\alpha,q_n}M_n^{-1/q_n}$ between different $O_{n,i}$, and $\rho(f_n(x), f_n(p_{n,i})) < \alpha/32$ for $x \in O_{n,i}$. Since finite-support boxes form a local base, choose bricks $B_{n,i} \subseteq O_{n,i}$ with $p_{n,i} \in \mathrm{int}\, B_{n,i}$, support level $R_n \ge K_n$, and $\mathrm{diam}_\rho(B_{n,i}) < \gamma_n$. As $B_{n,i} \subseteq J_n$ in coordinates $\ell \le K_n$, the intervals $I_\ell^{n,i}$ for $\ell \le N$ lie in the anchored site intervals, hence in $[2\theta, 1 - 2\theta]$: condition (S0) of Definition 3.2 holds, and (S1), (S2) and the window membership hold by construction.

Let $c_{n+1} = f_n(p_{n,1})$ and define the next anchored site $J_{n+1} = \prod_{\ell \le K_{n+1}}[u_\ell, u_\ell + \alpha] \times [0,1]^{\{K_{n+1}+1, \dots\}}$ with $u_\ell = \min\big(\max((c_{n+1})_\ell - \alpha/2,\, 2\theta),\, 1 - 2\theta - \alpha\big)$. Every point $y \in J_{n+1}$ satisfies $|y_\ell - (c_{n+1})_\ell| \le \alpha + 2\theta$ for $\ell \le K_{n+1}$, hence $\rho(y, c_{n+1}) \le \alpha + 2\theta + 2^{-K_{n+1}}$.

Define the model map $q_{n,i} \in C(Q_\infty, Q_\infty)$ coordinatewise. Writing $I_\ell^{n,i} = [b_\ell, c_\ell]$ and $\lambda_\ell(t) = \min\big(\max\big(\frac{t - b_\ell}{c_\ell - b_\ell}, 0\big), 1\big)$:

$$\pi_\ell q_{n,i}(x) = \begin{cases} (u_\ell - 2\theta) + (\alpha + 4\theta)\,\lambda_\ell(x_\ell), & \ell \le N, \\ u_\ell + \alpha/2, & N < \ell \le K_{n+1}, \\ (c_{n+1})_\ell, & \ell > K_{n+1}. \end{cases}$$

The values in the first line range over $[u_\ell - 2\theta,\, u_\ell + \alpha + 2\theta] \subseteq [0,1]$ by the anchoring of J_{n+1}. Choose Urysohn functions $\varphi_{n,i} : Q_\infty \to [0,1]$ with $\varphi_{n,i} = 1$ on $B_{n,i}$ and $\varphi_{n,i} = 0$ off $O_{n,i}$, and set

$$g_n = \Big(1 - \sum_i \varphi_{n,i}\Big) f_n + \sum_i \varphi_{n,i}\, q_{n,i},$$

a coordinatewise convex combination, hence a continuous self-map of Q_∞ agreeing with f_n off $\bigcup_i O_{n,i}$ and with $q_{n,i}$ on $B_{n,i}$.

Error estimate. For $x \in O_{n,i}$ and $\ell \le N$, the value $\pi_\ell q_{n,i}(x)$ lies in the 2θ-enlargement of the site interval $[u_\ell, u_\ell + \alpha]$, every point of which is within $\alpha + 4\theta$ of $(c_{n+1})_\ell$; for $N < \ell \le K_{n+1}$ the value is the site centre, within $\alpha/2 + 2\theta$ of $(c_{n+1})_\ell$; and for $\ell > K_{n+1}$ it equals $(c_{n+1})_\ell$ exactly. Hence

$$\rho\big(q_{n,i}(x), c_{n+1}\big) \le \sum_{\ell \le K_{n+1}} 2^{-\ell}\big(\alpha + 4\theta\big) \le \alpha + 4\theta.$$

Also $\rho(c_{n+1}, f_n(x)) \le \rho(f_n(p_{n,1}), f_n(p_{n,i})) + \rho(f_n(p_{n,i}), f_n(x)) < \alpha/16 + \alpha/32 < \alpha/8$. Since $4\theta = \alpha/2$, we get $\rho(g_n(x), f_n(x)) \le \rho(q_{n,i}(x), f_n(x)) < \alpha + \alpha/2 + \alpha/8 < 2\alpha$, and therefore $D(f, g) \le 2\alpha < \eta$.

Verification of (S3). The bricks at time $n+1$, constructed at the next step inside $\mathrm{int}(J_{n+1})$, have ℓ-intervals contained in $[u_\ell, u_\ell + \alpha]$ for every $\ell \le K_{n+1}$, so $m^-_{n+1,\ell} \ge u_\ell$ and $m^+_{n+1,\ell} \le u_\ell + \alpha$ for $\ell \le N$. If $x \in B_{n,i}$ has $x_\ell = b_\ell$, then $\pi_\ell g_n(x) = \pi_\ell q_{n,i}(x) = u_\ell - 2\theta \le m^-_{n+1,\ell} - 2\theta$, and on the opposite face the value is $u_\ell + \alpha + 2\theta \ge m^+_{n+1,\ell} + 2\theta$. Thus (S3) holds with margin $2\theta > \theta$, for *whatever* bricks the next step selects inside J_{n+1}. Carrying out the induction for all n yields $g \in \mathcal{U}_N(W_\bullet)$ with $D(f, g) < \eta$. □

5 The residual depth hierarchy

Theorem 5.1 (Finite-depth genericity)**.** *For every $N \ge 1$ and every window sequence $W_\bullet$ the class $\mathcal{U}_N(W_\bullet)$ is open and dense in $(\mathcal{F}(Q_\infty), D)$. Hence*

$$\mathcal{S}(W_\bullet) = \bigcap_{N \ge 1} \mathcal{U}_N(W_\bullet)$$

is residual. Moreover, for any countable family $\{W^a_\bullet\}_{a \ge 1}$ of window sequences, $\bigcap_{a \ge 1} \mathcal{S}(W^a_\bullet)$ is residual, and every g in this set admits, for every a and every depth N, a strict depth-N $W^a_\bullet$-windowed brick scheme with the exact one-step covering property of Theorem 3.4.

Proof. Openness is Lemma 3.3, density is Lemma 4.3, and $(\mathcal{F}(Q_\infty), D)$ is Baire by Lemma 2.1; countable intersections of residual sets are residual. The covering property is Theorem 3.4. □

6 Generic entropy bursts in every tail

The depth hierarchy of Theorem 5.1 is a statement about one-step structure. We now prove that a genuinely *dynamical*, infinite-horizon shadow of the dense theorem of [1] also holds residually. Recall the tail Bowen data: for $g \in \mathcal{F}(Q_\infty)$ and $r \ge 0$, $G^0_r = \mathrm{id}$, $G^j_r = g_{r+j} \circ \cdots \circ g_{r+1}$, the Bowen metric $\rho^g_{r,N}(x, y) = \max_{0 \le j < N} \rho(G^j_r(x), G^j_r(y))$, and $s_{r,N}(g, \varepsilon)$, the maximal cardinality of an (r, N, ε)-separated set. The tail scale entropy is $H_{g,r}(\varepsilon) = \limsup_N \frac{1}{N} \log s_{r,N}(g, \varepsilon)$.

Lemma 6.1. *For fixed $r \ge 0$, $N \ge 1$, $\varepsilon > 0$ and $m \in \mathbb{N}$, the set $\{g \in \mathcal{F}(Q_\infty) : s_{r,N}(g, \varepsilon) \ge m\}$ is open.*

Proof. Let g admit an (r, N, ε)-separated set $E = \{x_1, \dots, x_m\}$, with $\min_{u \ne v} \rho^g_{r,N}(x_u, x_v) = \varepsilon + 2\kappa$ for some $\kappa > 0$. The maps $g_{r+1}, \dots, g_{r+N-1}$ are uniformly continuous; hence there is $\delta > 0$ such that $D(h, g) < \delta$ implies $\sup_x \rho\big(H^j_r(x), G^j_r(x)\big) < \kappa$ for $0 \le j < N$, by a finite induction on j: $\rho(H^j_r(x), G^j_r(x)) \le \rho\big(h_{r+j}(H^{j-1}_r x), g_{r+j}(H^{j-1}_r x)\big) + \rho\big(g_{r+j}(H^{j-1}_r x), g_{r+j}(G^{j-1}_r x)\big) \le D(h, g) + \omega_{g_{r+j}}\big(\rho(H^{j-1}_r x, G^{j-1}_r x)\big)$, where ω_u is a modulus of continuity of u. Then E is (r, N, ε)-separated for every such h. □

Theorem 6.2 (Residual entropy bursts)**.** *The set*

$$\mathcal{R} \;=\; \Big\{ g \in \mathcal{F}(Q_\infty) :\ \forall r \ge 0\ \forall q \ge 1\ \forall j \ge 1 :\ \limsup_{N \to \infty} \sup_{0 < \varepsilon < 1/j} \frac{\log s_{r,N}(g, \varepsilon)}{N \log(1/\varepsilon)} \;\ge\; q \Big\}$$

is residual in $(\mathcal{F}(Q_\infty), D)$*. In particular, for a residual set of systems,* every *tail exhibits separated orbit growth of every order* q *at arbitrarily fine scales:* $\limsup_N \sup_{\varepsilon<1/j} \frac{\log s_{r,N}}{N\log(1/\varepsilon)} = \infty$ *for every* r *and* j*.*

Proof. For r, q, j, N_0 let

$$V_{N_0}^{r,q,j} = \bigcup_{\varepsilon\in\mathbb{Q},\ 0<\varepsilon<1/j} \ \bigcup_{N\geq N_0} \{\, g :\ s_{r,N}(g,\varepsilon) > \varepsilon^{-qN} \,\}.$$

Each $V_{N_0}^{r,q,j}$ is open by Lemma 6.1. We claim it is dense. Let $f \in \mathcal{F}(Q_\infty)$ and $\eta > 0$. By the dense Hilbert-cube theorem of [1, Theorem 3.6] there is g with $D(f,g) < \eta$ and constants $C_{g,q+1}$ such that $H_{g,r}(\varepsilon) \geq (q+1)\log(1/\varepsilon) - C_{g,q+1}$ for every $r \geq 0$ and all sufficiently small $\varepsilon > 0$. Choose a rational $\varepsilon < 1/j$ small enough that $(q+1)\log(1/\varepsilon) - C_{g,q+1} > q\log(1/\varepsilon)$, i.e. $\log(1/\varepsilon) > C_{g,q+1}$. Since $H_{g,r}(\varepsilon) > q\log(1/\varepsilon)$, there are arbitrarily large N with $\frac{1}{N}\log s_{r,N}(g,\varepsilon) > q\log(1/\varepsilon)$, in particular some $N \geq N_0$; then $s_{r,N}(g,\varepsilon) > \varepsilon^{-qN}$ and $g \in V_{N_0}^{r,q,j}$. Thus each $V_{N_0}^{r,q,j}$ is open dense, and $\mathcal{R} \supseteq \bigcap_{r,q,j,N_0} V_{N_0}^{r,q,j}$ is residual: indeed, if g lies in the intersection then for all r, q, j and every N_0 there are $N \geq N_0$ and $\varepsilon < 1/j$ with $\log s_{r,N}(g,\varepsilon) > qN\log(1/\varepsilon)$, which is exactly the displayed lim sup sup condition. □

Remark 6.3. *Theorem 6.2 is the residual shadow of the dense metric-mean-dimension theorem: it asserts unbounded* burst *complexity, where the certifying scale* ε *may vary with the horizon* N*. Upgrading it to residual infinite* upper metric mean dimension *— where a single scale sequence* $\varepsilon \downarrow 0$ *must work for fixed-*ε *horizon limits — runs precisely into the scale-pinning issue discussed in Section 1; we leave it as part of Problem 8.1. Note also that near contracting systems the fixed-scale sets* $\{g : H_{g,r}(\varepsilon_0) > 0\}$ *fail to be dense (a* δ*-ball around a constant system has all tail orbits eventually* 2δ*-confined, killing all separation at scales above* 4δ*), so the burst formulation, with its scale flexibility, is the natural residual statement at this level of generality.*

7 The obstruction: exact schemes have empty interior

We recall the exact class of [1].

Definition 7.1 (Exact windowed Hilbert horseshoe; [1, Definition 12.1])**.** *A system* $g \in \mathcal{F}(Q_\infty)$ *has an exact* $W_\bullet$*-windowed Hilbert horseshoe if there are* $\alpha \in (0, 1/4)$*, integers* $K_n \geq q_n$*,* (α, K_n)*-boxes* J_n*, and finite-support boxes* $B_{n,1}, \ldots, B_{n,M_n} \subseteq \mathrm{int}(J_n) \cap W_n$ *with* $\mathrm{diam}_\rho(B_{n,i}) < \gamma_n$*, pairwise separation* $\mathrm{dist}_{q_n}(B_{n,i}, B_{n,j}) > c_{\alpha,q_n} M_n^{-1/q_n}$*, and*

$$g_n(B_{n,i}) = J_{n+1} \qquad (n \geq 1,\ 1 \leq i \leq M_n).$$

Let Ex *denote the set of such systems.*

Theorem 7.2 (Empty interior)**.** Ex *is dense in* $(\mathcal{F}(Q_\infty), D)$ *and has empty interior. More generally, let* $\mathcal{P} \subseteq \mathcal{F}(Q_\infty)$ *be any class such that every* $g \in \mathcal{P}$ *admits some* $n \geq 1$ *and a compact set* $A \subseteq Q_\infty$ *for which* $\pi_j\big(g_n(A)\big) = [0,1]$ *for infinitely many coordinates* j*. Then* $\mathcal{P}$ *has empty interior.*

Proof. Density of Ex is [1, Proposition 12.2]. For the rest, fix any $g \in \mathcal{F}(Q_\infty)$ and $\delta > 0$; we produce h with $D(h,g) < \delta$ and $h \notin \mathcal{P}$. Choose L with $2^{-L} < \delta$ and define the *coordinate*

squeeze $\Phi : Q_\infty \to Q_\infty$,

$$\Phi(y)_j = \begin{cases} y_j, & j \le L, \\ \frac{1}{8} + \frac{3}{4}\, y_j, & j > L. \end{cases}$$

Φ is continuous and $\rho(\Phi(y), y) = \sum_{j>L} 2^{-j} \left|\frac{1}{8} - \frac{y_j}{4}\right| \le \frac{3}{8}\, 2^{-L} < \delta$. Set $h = (\Phi \circ g_n)_{n\ge 1}$, so $D(h, g) < \delta$. For every n and every $x \in Q_\infty$ and every $j > L$ we have $\pi_j h_n(x) \in [\frac{1}{8}, \frac{7}{8}]$. Hence for every n and every compact A, $\pi_j(h_n(A)) \subseteq [\frac{1}{8}, \frac{7}{8}] \neq [0, 1]$ for all $j > L$: the image projections of h are full in at most finitely many coordinates. Thus $h \notin \mathcal{P}$, and $\mathcal{P}$ contains no D-ball.

Finally, Ex is such a class: if $g \in \mathsf{Ex}$, then $g_1(B_{1,1}) = J_2$, and the (α, K_2)-box J_2 has $\pi_j(J_2) = [0, 1]$ for every $j > K_2$, which is an infinite set of coordinates. Hence $\mathrm{int}(\mathsf{Ex}) = \emptyset$. □

Remark 7.3. *Theorem 7.2 sharpens the observation of [1, Section 14] from "not asserted to be open" to "cannot meet any open set in a stable way": the dense class* Ex *is topologically thin in the strongest possible sense compatible with density. The squeeze perturbation also destroys surjectivity of each map and every property forcing deep full-range images, e.g. exactness onto any finite-support box family with unbounded supports. Together with Remark 3.5, this closes the open-versus-dense question raised in [1, Section 14]: openness is impossible at infinite depth, holds with radius* $\theta 2^{-N-1}$ *at depth* N*, and the residual object produced by Baire fusion is the full depth hierarchy* $\mathcal{S}(W_\bullet)$ *of Theorem 5.1.*

8 What remains open

Problem 8.1. *Let* $W_\bullet$ *be a sequence of dense open subsets of* Q_∞ *and consider the class* $\mathcal{C}(W_\bullet)$ *of systems* $g \in \mathcal{F}(Q_\infty)$ *such that for every* $r \ge 0$ *the tail* $\mathrm{tail}_r\, g$ *admits a compact symbolic semi-factor onto a full variable-alphabet shift, a continuum-sized DC1-scrambled set visiting* W_{r+j+1} *at tail time* j*, infinite topological entropy, and* $H_{g,r}(\varepsilon) \ge q \log(1/\varepsilon) - C_{g,q}$ *for every finite* q*. By [1, Theorem 3.6],* $\mathcal{C}(W_\bullet)$ *is dense. Is* $\mathcal{C}(W_\bullet)$ *residual? Is the subclass with infinite upper metric mean dimension in every tail residual?*

The results of this paper bound the possible strategies. By Theorem 7.2, no proof can proceed through an open class of systems with unbounded-depth exact or margin-protected covering, since every such class has empty interior; by Theorem 5.1 the natural residual envelope is the depth hierarchy $\mathcal{S}(W_\bullet)$, on which one-step covering is exact at every finite depth but infinite-horizon orbit selection is not automatic; and by Theorem 6.2 the burst form of infinite-order complexity is already residual. What is missing is exactly a mechanism producing, for a residual set of systems, orbits threading infinitely many prescribed bricks — equivalently, a replacement for the backward induction of [1, Lemma 13.1] that consumes only finite-depth information per step. The quantitative erosion estimate of Remark 3.5 suggests that such a mechanism, if it exists, must interleave depth and horizon nontrivially; we intend to return to this elsewhere.

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Declaration on generative AI

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